Predicting Malaria Cases in Benin Using Climate Data and Machine Learning

A Data-Driven Research Proposal

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Abstract

Malaria remains one of the most serious public health challenges in Benin, affecting thousands of people each year. Accurately predicting future cases is essential for improving prevention strategies and healthcare planning.

This project aims to predict malaria incidence in Benin by combining historical epidemiological data with climate variables such as rainfall and temperature. Machine learning models, including Random Forest and Linear Regression, are used to identify patterns and generate reliable predictions.

The datasets are collected, cleaned, and merged to build a predictive model. The model is then evaluated using standard error metrics and applied to forecast malaria cases over the coming years.

The results are presented through an interactive dashboard, allowing users to explore trends and predictions in a simple and intuitive way. This project demonstrates how data science and artificial intelligence can support informed decision-making in public health.

Research Background

Malaria is a major public health challenge in many African countries, including Benin. It is transmitted by mosquitoes and is strongly influenced by environmental factors such as rainfall and temperature. These conditions directly affect mosquito breeding and survival, which in turn influence the spread of the disease.

In Benin, seasonal variations—especially during the rainy season—are often associated with an increase in malaria cases. However, accurately predicting these fluctuations remains a challenge. Traditional approaches mainly rely on historical observations and do not fully take advantage of modern data analysis techniques.

With the advancement of data science, it is now possible to analyze large datasets and uncover hidden patterns. Machine learning models provide powerful tools to predict

future malaria trends based on both historical and environmental data.

This research applies these techniques to better understand and anticipate malaria patterns in Benin. The goal is to provide useful insights that can support public health planning, improve resource allocation, and enable earlier intervention strategies.

Research Questions

This study is guided by the following research questions:

1. Can malaria cases in Benin be accurately predicted using climate data and historical records?
2. What is the relationship between rainfall and malaria incidence?
3. How does temperature influence the spread of malaria?
4. To what extent do previous malaria cases contribute to future outbreaks?
5. How accurate and reliable are machine learning models in predicting malaria trends?

Methodology

This research adopts a data-driven approach to predict malaria cases in Benin. The methodology is structured into the following steps:

1. Data Collection
   - Malaria data is obtained from publicly available sources such as Our World in Data. Climate data, including rainfall and temperature, is collected for the same time period to ensure consistency.

2. Data Preparation
   - The datasets are cleaned and structured to ensure accuracy and consistency. Only data related to Benin is retained. The malaria and climate datasets are then merged based on the year.
   - Additionally, a new feature called “previous year cases” is created to capture temporal trends in malaria incidence.

3. Feature Selection
   - Input variables: Rainfall, Temperature, Previous year malaria cases, Year
   - Target variable: Malaria cases

4. Model Development
   - Linear Regression (baseline model)
   - Random Forest Regressor (advanced model)

5. Model Evaluation
   - Metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE)

6. Prediction
   - Forecast malaria cases for future years based on input variables

7. Visualization
   - Graphical analysis and interactive dashboard (Streamlit)

Expected Results

This study is expected to produce a machine learning model capable of predicting malaria cases in Benin with a satisfactory level of accuracy.

The results are expected to reveal a meaningful relationship between climate factors—such as rainfall and temperature—and malaria incidence. Previous year malaria cases are anticipated to play a significant role in improving prediction performance.

The model will generate forecasts for future malaria cases, which will be presented through graphs and tables. These visualizations will facilitate a clearer understanding of trends and patterns over time.

Overall, this study is expected to demonstrate that machine learning can serve as an effective tool for analyzing and predicting malaria trends using environmental and historical data.

Significance

This research is significant because it applies data science and machine learning to address a critical public health issue.

The findings can support better decision-making in malaria prevention by providing insights into how climate factors influence disease transmission. This can assist health authorities in planning interventions, allocating resources more efficiently, and preparing for potential outbreaks.

Furthermore, this project highlights the practical application of artificial intelligence in epidemiology, demonstrating its potential to address healthcare challenges, particularly in developing countries.

Finally, this research contributes to academic and professional development by showcasing skills in data analysis, machine learning, and problem-solving, which are essential in modern scientific and technological fields.

Timeline

Phase 1: Data Collection — 1 week
Phase 2: Data Cleaning and Preparation — 1 week
Phase 3: Model Development — 2 weeks
Phase 4: Model Evaluation — 1 week
Phase 5: Visualization and Dashboard Development — 1 week
Phase 6: Writing and Final Review — 1 week

Conclusion

In conclusion, this study demonstrates the potential of machine learning techniques to predict malaria cases in Benin using climate and historical data.

By identifying key relationships between environmental factors and malaria incidence, the project provides valuable insights that can support public health decision-making.

Although the model has certain limitations, it represents an important step toward integrating data science into disease prediction and prevention strategies.

Future improvements could include the use of additional variables, more advanced models, and real-time data integration to enhance prediction accuracy and practical application.

Limitations

This study has some limitations that should be considered.

First, the accuracy of the model depends on the quality and availability of data. Limited or incomplete data may affect prediction performance.

Second, the model uses a limited number of variables (rainfall, temperature, and previous cases), while other factors such as healthcare access, population density, and interventions may also influence malaria transmission.

Finally, the predictions are based on historical patterns and may not fully capture unexpected changes such as climate anomalies or policy interventions.

Despite these limitations, the model provides a useful foundation for understanding malaria trends.

Future Work

Future work can improve this study in several ways.

Additional data sources, such as humidity, population density, and healthcare infrastructure, could be included to enhance model performance.

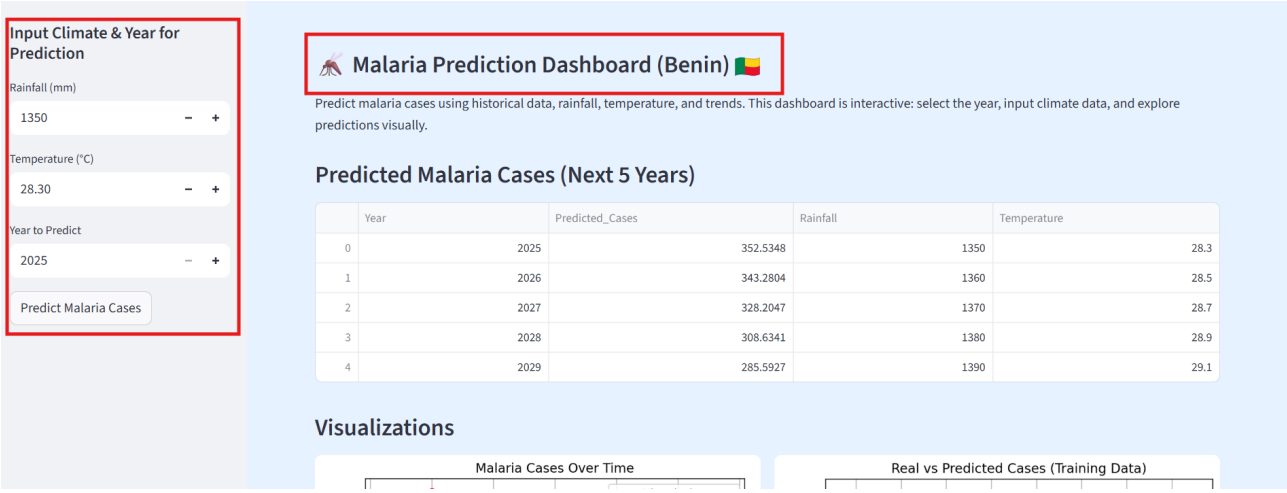
More advanced machine learning techniques, such as deep learning models, could also be explored for improved prediction accuracy.

Furthermore, integrating real-time data and deploying the model as a live monitoring system could provide continuous updates for health authorities.

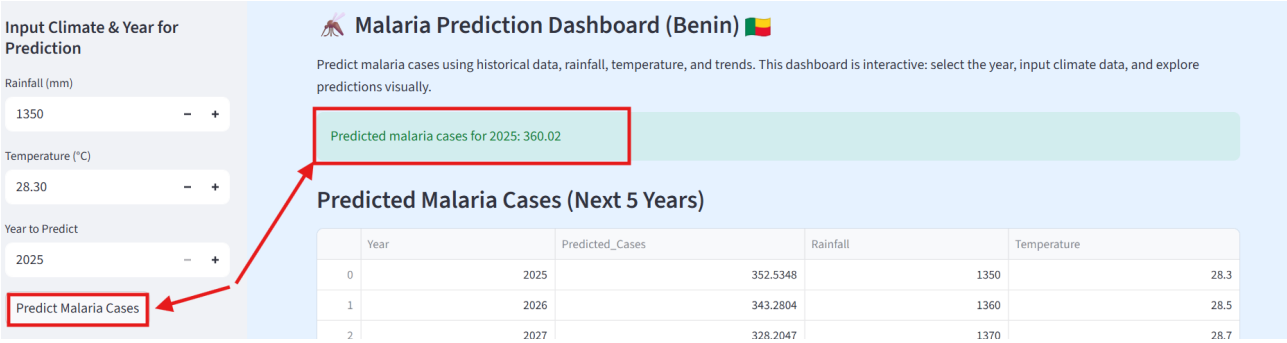
These improvements would make the system more robust and applicable in real-world public health scenarios.

Application Preview

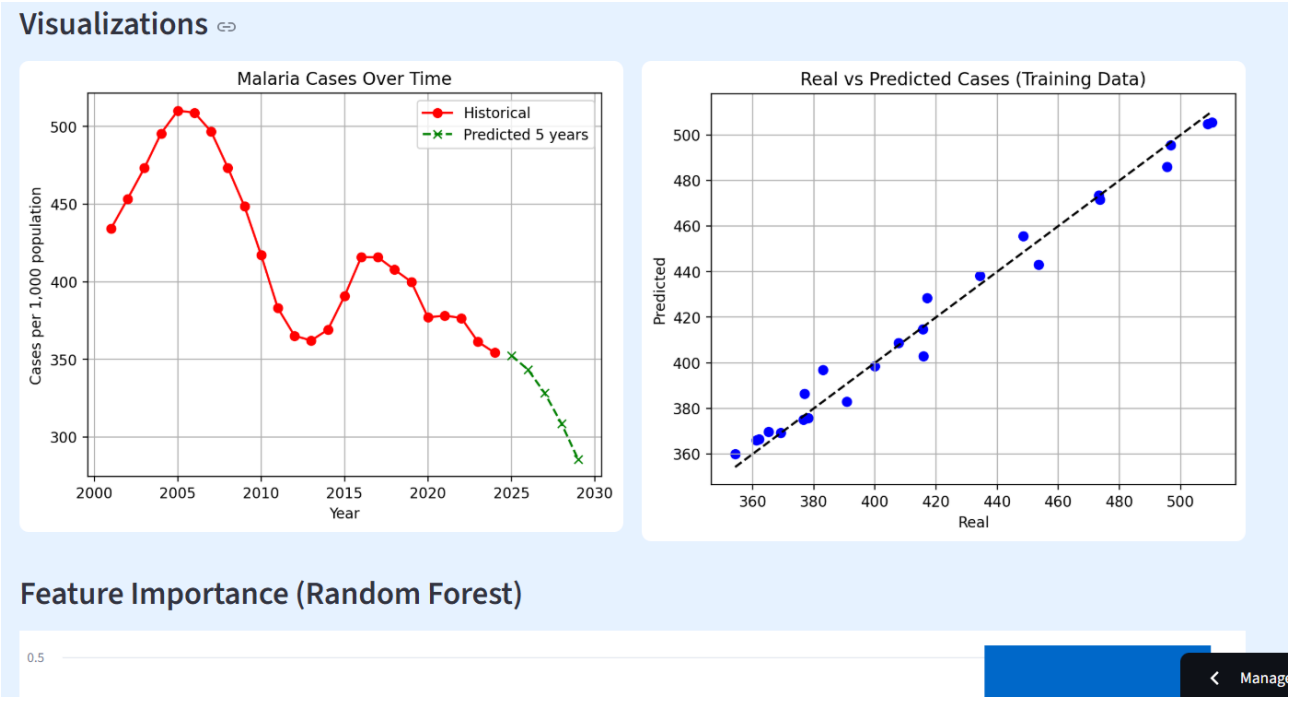
Dashboard Overview



Prediction Result



Data Visualization



Live Application

The interactive dashboard for this project is available online:

https://malaria-prediction-benin-fadil-ade.streamlit.app/

Users can explore malaria trends, input climate data, and generate predictions directly through the web application.

Bibliography

- Our World in Data. Malaria Dataset. Available at: https://ourworldindata.org/
- Climate Data Sources (Rainfall and Temperature Datasets)
- Scikit-learn Documentation. Available at: https://scikit-learn.org/
- Python Documentation. Available at: https://docs.python.org/

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